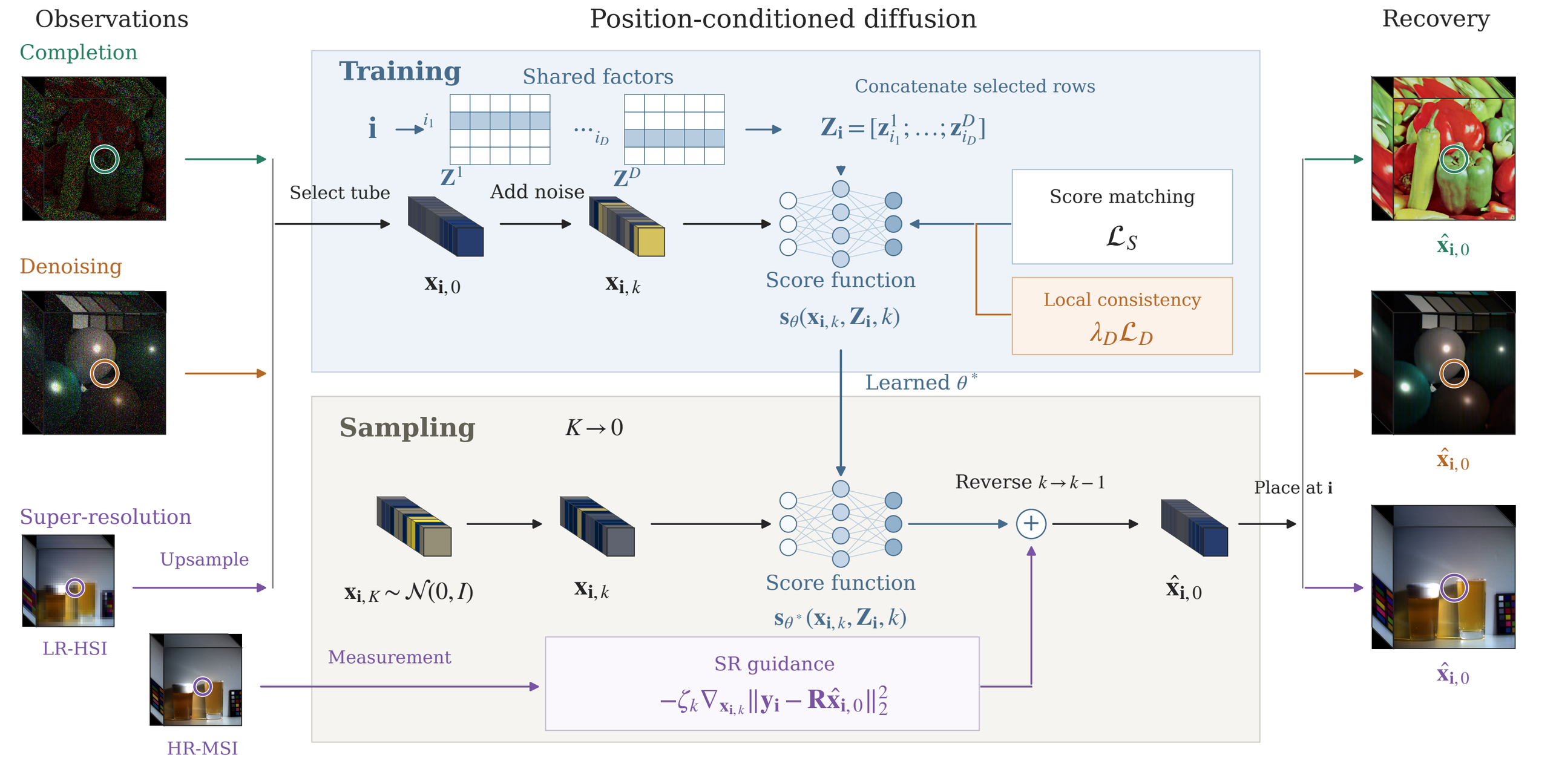

Observations
Position-conditioned diffusion
Recovery
Completion
Denoising
Super-resolution
Upsample
LR-HSI
HR-MSI
Training
Shared factors
$\mathbf{i} \to i_1 \cdots i_D$
$\mathbf{Z}^1$
$\mathbf{Z}^D$
Concatenate selected rows
$\mathbf{Z}_\mathbf{i} = [\mathbf{z}^1_{i_1}; \ldots; \mathbf{z}^D_{i_D}]$
Select tube
Add noise
$\mathbf{x}_{\mathbf{i},0}$
$\mathbf{x}_{\mathbf{i},k}$
Score function
$\mathbf{s}_\theta(\mathbf{x}_{\mathbf{i},k}, \mathbf{Z}_\mathbf{i}, k)$
Score matching
$\mathcal{L}_S$
Local consistency
$\lambda_D \mathcal{L}_D$
Learned θ^*
Sampling
$K \to 0$
$\mathbf{x}_{\mathbf{i},K} \sim \mathcal{N}(0, I)$
$\mathbf{x}_{\mathbf{i},k}$
Score function
$\mathbf{s}_{\theta^*}(\mathbf{x}_{\mathbf{i},k}, \mathbf{Z}_\mathbf{i}, k)$
Reverse $k \to k-1$
$\hat{\mathbf{x}}_{\mathbf{i},0}$
Measurement
SR guidance
$-\zeta_k \nabla_{\mathbf{x}_{\mathbf{i},k}} \|\mathbf{y}_\mathbf{i} - \mathbf{R}\hat{\mathbf{x}}_{\mathbf{i},0}\|_2^2$
Place at $\mathbf{i}$
$\hat{\mathbf{x}}_{\mathbf{i},0}$
$\hat{\mathbf{x}}_{\mathbf{i},0}$
$\hat{\mathbf{x}}_{\mathbf{i},0}$